

PhD Project Outline

The Value of Information in Private-Market Investment Decisions

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Target institution

University of St. Gallen — Chair of Private Markets & Alternative Investments, School of Finance (Prof. Tereza Tykvová), Swiss Finance Institute. A methodological co-advisor at ETH (decision-making under uncertainty / machine learning) is envisaged for the modelling side.

1. Research context and focus

Investors in private markets decide under extreme uncertainty on fragmentary, non-standardised information. Much empirical work — including your own — establishes *what* drives private-market outcomes (syndication, selection, the risk profile of buyouts). The question I want to open sits one step earlier and is, so far, largely unasked: given a decision, *which* missing piece of information — at what cost — most improves it? Not which factor predicts success, but which piece of diligence is worth acquiring.

This builds on my accepted WI2026 Student Track paper (*Can Non-Financial Signals Price Private Companies?*), which shows on 3,403 PitchBook deals, under a strict out-of-time holdout, that non-financial signals price private companies competitively with financial baselines — and that **investor-syndicate structure**, not firm financials, is the dominant driver. That last finding connects directly to your work on private-equity and venture syndication, and it exposes the ceiling of a purely predictive approach.

2. Research questions

Main question: In private-market investment decisions, which information acquisition — at what cost — most improves the quality of the decision?

- How can each decision be represented by its *information state*: what is knowable at time t , reconstructed point-in-time, before the outcome is realised?
- How do we quantify the *marginal value* of an additional piece of information relative to its cost?
- Does the value of information differ across settings — venture vs. buyout, syndicated vs. solo, primary vs. secondary deals?

3. Theoretical framework

The project brings the economics of *value of information* (Howard; Savage) into private-market investment decisions. Your empirical private-markets programme identifies the factors that matter for outcomes; value-of-information theory asks the normative counterpart — *what is worth knowing next, and whether it is worth its cost*. The core object is the expected improvement in decision quality from moving $\text{Decision}(\text{state}) \rightarrow \text{acquire info (cost } c) \rightarrow \text{state}' \rightarrow \text{Decision}(\text{state}')$. The technical machinery (Bayesian experimental design; sequential decision-making under uncertainty) exists in the ML literature; its application to private-market diligence is open, and it needs a rigorous empirical-finance anchor to be posed and tested credibly.

4. Methodology

Feasibility does not depend on proprietary data. Ground truth comes from a *point-in-time reconstruction* of public sources (so the information state at time t excludes look-ahead — an identification strategy familiar from empirical finance) plus a *virtual-fund simulator* that measures decision quality counterfactually. On top of this I estimate the value of each information type and rank diligence steps by expected value per unit cost, benchmarked against the WI2026 pipeline as a predictive baseline.

- **Year 1:** formalise the information-state / value-of-information framing; build the point-in-time dataset and simulator.
- **Year 2:** estimate information value across settings; complete and submit the flagship paper.
- **Year 3–4:** extensions (syndication signals, buyout vs. venture), writing, publications.

5. Expected contribution

The project contributes (i) a *value-of-information framing for private-market decisions* — a normative account of which signals are worth acquiring; (ii) a reproducible point-in-time and simulator methodology that makes the question testable without proprietary data; and (iii) empirical evidence on where diligence effort is mis-allocated across deal types. It speaks directly to entrepreneurial-finance and private-equity audiences and generalises across the alternative-investments spectrum, which makes it a research line rather than a single study.

6. Fit with host institution

Your Chair — empirical private markets, entrepreneurial finance, and alternative investments, with a strong record on syndication, buyouts, and selection — is exactly the domain this question needs to be posed rigorously. Your survey of the VC/PE literature and its research agenda maps the very gaps a value-of-information perspective can address, and your work on syndication connects to my paper's central finding. I would value your Chair as the empirical-finance home for the project, with an ETH methodological co-advisor for the modelling. I already have an accepted publication in the area and want to build a coherent line of research, not a single paper.

7. Selected references

- Howard, R. A. (1966). *Information Value Theory*. IEEE Trans. Systems Science and Cybernetics, 2(1), 22–26.
- Savage, L. J. (1954). *The Foundations of Statistics*. Wiley.
- Tykvová, T. (2018). *Venture Capital and Private Equity Financing: An Overview of Recent Literature and an Agenda for Future Research*. (SSRN working paper / survey.)
- Pursiainen, V., & Tykvová, T. *Retail Customer Reactions to Private Equity Acquisitions*. (working paper.)
- Guidi, A., Rashid, S., & Zhong, H. (2026). *Can Non-Financial Signals Price Private Companies?* WI2026 Student Track (accepted).

One-page WI2026 summary attached. Code and manuscript available on request.